

Application Architecture Review Request

AlphaPaaS Cloud Broker — Architecture Governance

Form reference	AR-2026-047
Submitted	14 April 2026
Requester	Léa Moreau, QuoteCraft Product Team
Assigned reviewer	(to be filled in by Cloud Broker)
Target completion	End of May 2026

1. Application overview

Application name	QuoteCraft
Application ID	AI-8742
Business owner	Léa Moreau
Technical owner	Tomás Ferrer
Cloud Broker contact	Hendrik Vos
Status	In production since November 2025

QuoteCraft is an internal microservice that produces price quotations for motor and home insurance policies. It is called by the broker-facing portal and by two partner APIs. It processes 180,000 to 220,000 quote requests per business day, with pronounced peaks on Monday mornings and on the last two working days of each month. This is the first annual architecture review since go-live.

2. Criticality and compliance

Criticality tier	Silver (per AlphaPaaS Availability & Resilience Standard v3.2)
Regulatory scope	GDPR; local insurance regulator reporting
Data classification	PII + Financial (Restricted)
Data residency	European Union only
Retention	Quote records: 7 years. Diagnostic logs: 30 days.

3. Technical profile

Application type	Stateless HTTP microservice — Python 3.11, FastAPI
Target hosting platform	AlphaPaaS on AWS (eu-central-1)
Application zone	ai-prod-int-01 (internal exposure)
Primary data store	Azure Database for PostgreSQL, Flexible Server
Shared file storage	AWS EFS (policy rate tables, PDF templates)
Cache	AWS ElastiCache for Redis

Secrets management	AWS Secrets Manager and Azure Key Vault, per local CSP
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External dependencies: the AlphaPaaS internal Pricing API (HTTPS) and the Atlas Credit Bureau API (third-party, HTTPS).

4. Exposure and authentication

Exposure	Internal only — intended router label: router=irp
Broker portal	mTLS
Partner A	OAuth 2.0 (client credentials)
Partner B	API key (legacy; migration to OAuth planned for Q4 2026)
WAF	Not applicable (no public exposure)

5. Expected load

Baseline throughput	50 requests / second
Peak throughput	400 requests / second
Forecast growth	2× over the next 18 months
Payload size	~12 KB request / ~40 KB response

6. Availability and disaster recovery

SLA commitment	99.5% monthly (silver tier)
RTO	4 hours
RPO	1 hour
Multi-AZ deployment	Required (silver tier)
DR testing	Not yet performed. Scheduled for Q3 2026.

7. Change history and known concerns

Initial launch was accelerated to meet the broker-portal go-live date.

Group Security flagged secrets handling as a concern in a spot-check in February 2026; remediation has not been completed.

The FinOps team flagged in March 2026 that QuoteCraft's non-production environment is consuming approximately 35% of what production consumes. The cause has not been investigated.

8. Review scope requested

Full architecture review across four dimensions:

- Cost — alignment with the FinOps standard, oversized resources, waste
- Security — OCP CIS compliance, secrets, network policies, encryption
- Scalability — whether the service can handle the forecast load
- Availability — whether the deployment meets silver tier requirements

Desired output: a findings report with severity ratings and concrete remediation steps. Actionable items are expected ready to be created as new backlog tickets, referencing this review.

9. Attachments and references

- Source repository — URL provided in the materials pack
- Architecture diagram — docs/architecture.png in the repository
- OpenAPI specification — docs/openapi.yaml in the repository
- Capacity plan (draft) — docs/capacity-plan.md in the repository
- Relevant policies — AlphaPaaS policy library, delivered separately as PDF